

MANTHAN BHALA

Machine Learning Engineer

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EXPERIENCE

Senior Machine Learning Engineer — Makro PRO

Oct 2023 – Present

Remote

- Built recommendation models: collaborative filtering, two-tower networks, and embeddings for 1M+ DAU.
- Increased basket adds by 52% through personalized user-based and product-based recommendations.
- Built multilingual (English, Thai) semantic product retrieval over 100k+ products with sub-100ms p99 latency.
- Built e-commerce image search combining SigLIP 2 image and text embeddings for cross-modal retrieval.
- Built personalization pipeline via feature engineering and LLM distillation into LightGBM.
- Created dynamic user segmentation for business visualization, product collection generation, and cold-start.
- Created a content generation POC using fine-tuned LLaMA 3 for multilingual titles, categories, descriptions.
- Developed end-to-end search and recommendation pipelines across retrieval, ranking, and reranking.
- Developed query suggestion and related-search models to improve customer search experience.
- Improved multilingual search via autocategorization, autocorrect, and query expansion.
- Built a multiuse RAG chatbot with FAQ handling, tool calling (search, order APIs), and guardrails.
- Built repeat purchase recommendations to predict reorder time and probability via fine-tuned LightGBM.

Machine Learning Engineer — Frinks AI

Mar 2023 – Jul 2023

Bengaluru, Karnataka, India

- Developed real-time ML pipelines for defect detection and item counting on manufacturing lines.
- Built computer vision and backend services for quality inspection, dimensioning, and crack detection.

Data Scientist — Jaguar Land Rover

Sep 2020 – Feb 2023

Bengaluru, Karnataka, India

- Developed ML pipelines and backend services for the Vehicle Algorithmic Self-Diagnostics Tool.
- Worked on ADAS systems enabling autonomous driving and driver-assistance features for vehicles.

Machine Learning Developer — Quantiphi Analytics

May 2019 – Jul 2019

Intern

- Developed and deployed ML models for video action recognition using CNN + LSTM.

SKILLS

ML Engineering: Python · PyTorch · TensorFlow · Hugging Face · LightGBM · PEFT

Applied AI: LLMs · Recommendation Systems · Information Retrieval · Learning to Rank · RAG

Data & Infrastructure: SQL · Docker · Kubernetes · CI/CD · Redis · vLLM · Vector Database · Typesense

EDUCATION

Bachelor of Technology — Indian Institute of Technology (IIT), Bombay

2016 – 2020 | CGPA: 8.48/10

PATENTS

2501007631 — A Hybrid Lexical-Semantic Retrieval Process for E-Commerce Search

2501004476 — A Hybrid LLM Process for Multi-Objective Reranking in E-Commerce

2501004419 — A Multilingual Phonetic BK-Tree Based Spelling Correction Process in E-Commerce